HIGH TIDE · INFORMATION BANK & DEFENCE BRIEF

# Give the bloom a way out.

Everything the team needs to know about the project, the site, the animal, the science, the engineering, the numbers and the world around them — plus every hard question we expect from the judges, with our best answer and the honest limit of each claim.

VERSION 2.0 · 11 NOVEMBER 2026 · INTERNAL — FOR THE TEAM AND MENTORS

HOW TO USE THIS DOCUMENT

# Read the summary. Memorise page 3. Rehearse chapter 20.

This is a reference, not a pitch. It is written so any team member can answer any question from a judge without guessing. Every number is traceable to a published source, to an assumption we chose and can defend, or to our own model.

| TAG | MEANING | HOW TO SAY IT OUT LOUD |
|---|---|---|
| SRC | Published or ENEC-stated figure, with a reference. | "The published figure is…" |
| ASM | An engineering assumption we chose. Reasonable, stated, replaceable by ENEC data. | "We assumed … because … ENEC data would replace it." |
| MOD | Output of our High Tide v1.0 transport model or design script. | "In our model…" — never "we showed". |
| HYP | A hypothesis we propose to test. Not evidence. | "Our hypothesis is… the first experiment tests it." |

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# Steer the bloom, don't fight it.

ENEC already detects jellyfish blooms offshore. The challenge asks what to do *after* the alert: how to keep blooms away from Barakah's cooling-water intakes with less manual work and without harming the sea. High Tide is a three-pronged answer that moves living jellyfish out of the intake's reach and verifies they stayed out.

### Prong 1
### Bubble corridor.

Segmented, controlled bubble curtains that throw surface water sideways at 0.30–0.43 m/s (≈10× swim speed). Only the 50 m sections near tracked animals fire; the wall mirrors when the tide turns.

### Prong 2
### The stick: (E)-2-nonenal.

A defence chemical released by injured jellyfish that deters a predatory jellyfish. Hypothesis: a thin pulsed band makes animals swim away from the intake.

### Prong 3
### The carrot: Artemia water.

Water conditioned by live brine shrimp smells like food. Medusae prefer that side of a tank. Released only at the exit the controller is using.

## The insight that makes it work

Intake water speeds up as Barakah's basin narrows. At the basin mouth (~4–6 km from the pumps) the mean pull is about **3 cm/s** [MOD]; a blue blubber swims at about **3.6 cm/s** [SRC]. Roughly **4.3 km** out, the two are equal. Inside that line every animal is on a one-way trip; outside it, a jellyfish that "decides" to leave still can. So High Tide puts its gate at the mouth, where bubbles are overwhelming and chemical nudges can still matter, and keeps a guard line and the existing screens as the backstop.

## What we can show today

- **A working transport model** (High Tide v1.0): 11 scenarios × 5 configurations × 3 seeds = 165 archived runs. With a mid-event tide reversal, controlled sections cut intake entries from 338 to 78 of 900 arrivals (seed 42) [MOD].
- **A cue layer, explicitly hypothetical**, capped at the animal's own swim speed: in the same reversal, intake entries fall a further 86 % (86 → 12, 3-seed mean); in weak currents, clearance rises 7× and compressor use halves [MOD] [HYP].
- **A site concept** traced on the real Barakah geometry: four intakes, a two-line 4.2 km gate across the basin mouth, two exits, cue stations, sensors, one shore station. All quantities sized (chapter 15).
- **A 26-week plan** with 11 deliverables and 5 go/no-go gates that each carry a number.

## What we are honest about

THE LIMITS — SAY THESE BEFORE THE JUDGES DO

- No one has shown an adult blue blubber swimming *away* from nonenal or *toward* Artemia water in open water. That is our first experiment.
- Our model is depth-averaged and uncalibrated; it is a design tool for comparing strategies, not a prediction of tonnes.
- The site geometry is traced from public imagery; the gate position, depths and currents need ENEC survey data.
- The first marine trial releases **zero** chemicals. Cues stay in contained lab work until four gates are passed.

## The ask

Bloom-load records at the screens, current data at the basin mouth, an introduction to a contained marine-lab partner, and a non-operational area to test one recoverable 50 m curtain section.

02 · NUMBERS TO MEMORISE

# The cheat sheet.

| TOPIC | NUMBER | CONTEXT | TAG |
|---|---|---|---|
| Peak-day load | **≈200 t/day** | Historic jellyfish load cited by ENEC mentors, 30 Oct 2026 | SRC |
| Animals on that day | **50,000** (25k–100k) | At 4 kg (2–8 kg) per 30–45 cm animal | ASM |
| Mean / peak hourly load | **8.3 / 25 t/h** | Peak = ×3 for tidal bunching; 6,250 animals/h, 1.7 per second | ASM |
| Per intake | **50 t/day** | ≈12,500 animals per intake per peak day | ASM |
| Cooling flow | **690 m³/s** | Whole-plant design maximum; 59.6 million m³/day | SRC |
| Plant | **4 × APR-1400, 5,600 MW** | ≈40 TWh/yr, up to 25 % of UAE electricity; all four in commercial operation since Sept 2024 | SRC |
| Breakwaters | **≈15 km** | Protect intake, discharge, navigation channel, wharf | SRC |
| Dredged depth | **5.5–9.5 m** below MSL | NMDC marine works 2011–2016 | SRC |
| Sea temperature at Barakah | **21.6 / 32.7 °C** | Winter / summer mean SST 2003–2019; salinity 44–45 PSU | SRC |
| Mean coastal current | **≈0.05 m/s** | Depth-averaged, ADOOS Buoy 9; M2 tidal amplitude 4.3 cm/s | SRC |
| Swim speed | **36 mm/s** | Small (2–5 cm) C. mosaicus in lab; adults untested | SRC |
| Pull at basin mouth / channel | **3 / 27 cm/s** | 690 m³/s ÷ traced width × assumed depth | MOD |
| Point of no return | **4.26 km** from intakes | Where mean pull = swim speed | MOD |
| Bubble surface current | **0.30–0.43 m/s** | Bulson relation at 6 m, 0.55–1.67 L/s per metre | MOD |
| Gate | **4.2 km span · 4,366 + 4,542 m** | Two mirrored lines, 179 sections of 50 m | ASM |
| Guard | **2 × 375 m** | Double curtain at channel mouth, 3.8 L/s·m, fires on breach | ASM |
| All hose / feeders | **9,657 m / 20,359 m** | 194 sections total | ASM |
| Compressors | **5 × 250 kW** (N+1) | 913 kW worst case = 0.016 % of plant output | MOD |
| Energy per event | **5.8 MWh** | 12 h event, ≈AED 1,700 | MOD |
| Stick band | **5 µg/L · 648 g/h · 7.8 kg/event** | Design envelope; 0.26 µg/L at the pumps | ASM |
| Stick safety margin | **1,092× / 21,000×** | Band / pumps vs 5.5 mg/L fish $LC_{50}$ (ECOSAR estimate) | MOD |
| Carrot | **0.5–5.4 m³/h** | From 5–54 kg live Artemia culture | ASM |
| Model headline | **338 → 78 → 8** | Intake entries of 900: none → controlled → +cues (reversal, seed 42) | MOD |
| CAPEX full scale | **AED 18–67 M** | Class-5 estimate | ASM |
| One unit-day | **33,600 MWh ≈ AED 6.2–9.9 M** | At USD 50–80/MWh | ASM |
| Payback | **2–11 unit-days** avoided | Insurance logic, not a claimed saving | MOD |
| Pilot ladder | **AED 2.2–8.6k → 36–125k → 106–486k** | Bench → contained biology → one marine trial | ASM |

# After the alert.

Advanced Technology Pioneers 2026 is run by the Advanced Technology Research Council (ATRC) with the Ministry of Education. Three sponsors set real problems: DCT Abu Dhabi, EDGE Group and ENEC. Finalists present at the NSTI Festival in November 2026; one winner per track.

## What ENEC actually asked

The ENEC statement is about the **sustainable management of seasonal jellyfish blooms** at coastal infrastructure. ENEC already runs a programme that detects blooms offshore and gives time to prepare. The challenge is explicitly **distinct from early detection**: it is about what to do once the early-warning notification has been issued. Solutions may **guide, redirect, collect, move or release** jellyfish, while **minimising manual intervention** and **protecting marine ecosystems**. Guiding examples: marine robotics, environmentally responsible guide/collect/relocate approaches, passive or active movement influence, and tools for situational awareness and integration with existing detection.

HOW HIGH TIDE MAPS TO THE BRIEF

| BRIEF ASKS FOR | HIGH TIDE ANSWER |
|---|---|
| Post-alert management | Starts at ENEC's alert; the controller arms on the alert and tracks arrivals. |
| Guide / redirect | Bubble gate routes animals to two exits; cues bias their own swimming. |
| Minimise manual intervention | Autonomous section control; no boats or crews in normal operation. |
| Protect marine ecosystems | No contact, no killing; zero chemical release until contained tests pass; non-target screening built into the plan. |
| Practical, developable | Commercial hardware (hose, compressors, ADCPs, cameras); staged pilot ladder; numeric gates. |
| Integration with detection | Uses ENEC's alert as the trigger; adds tracking and verification counts that can feed back to ENEC. |

## What ENEC said it will not give

The brief notes participants will not receive detailed facility design or internal operating data. Hence our approach: state assumptions clearly, trace public geometry, and make the first deliverable a precise data request.

## Timeline

| PHASE | DATES |
|---|---|
| Applications | 20 Aug – 11 Oct 2026 |
| Shortlist announced | 25 Oct 2026 |
| Mentoring (incl. 30 Oct ENEC session where the 200 t figure was cited) | 26 Oct – Nov 2026 |
| Demo Day / NSTI Festival | November 2026 |

# Four units, one basin, a very warm sea.

## The plant

| ITEM | VALUE | SOURCE |
|---|---|---|
| Location | Al Dhafra region, ~23.97° N, 52.23° E; ~220 km west of Abu Dhabi city (straight line) | Public coordinates |
| Reactors | 4 × APR-1400 PWR (Korean design; reference plant Shin Kori 3&4), 1,400 MW each, 5,600 MW total | ENEC; WNN |
| Output | ~40 TWh/yr; up to 25 % of UAE electricity | ENEC / WAM, Sept 2024 |
| Commercial operation | Unit 1 Apr 2021 · Unit 2 2022 · Unit 3 2023 · Unit 4 5 Sept 2024 | ENEC; NEI |
| Operator / owners | Nawah Energy Company (operations; now ENEC Operations), Barakah One Company (commercial), ENEC; regulator FANR | ENEC |
| Cooling | Once-through seawater (no towers); "ultimate heat sink" is the Gulf; water pumped through an intake channel to the condensers and returned | ENEC final cooling page |
| Design flow | Up to 690 $m^3/s$ whole plant (≈172 $m^3/s$ per unit if shared evenly — our arithmetic) | Nesterov et al. 2023 |
| Gulf adaptations | Larger pumps, heat exchangers and pipes; "more refined intake screen design"; modified breakwater to separate intake and discharge and avoid recirculation | ENEC FAQ; power-technology |
| Intake structures | EPC by HILALCO; reinforced concrete structures feeding condensers and heat exchangers | NS Energy |
| Marine works | NMDC 2011–2016: dredged intake channel, discharge channel, navigation channel; breakwaters; ~2.35 $km^2$ dredged, >9 million $m^3$ removed; bed 5.5–9.5 m below MSL | NMDC |
| Breakwaters | Quarry rock and concrete, ~15 km combined; built from early 2011; artificial reef added 2014 with EAD | ENEC news |

## The layout we traced

From public satellite imagery (Esri World Imagery, traced by the team): a long curved **west breakwater** runs from a head at about 52.191° E, 24.002° N, curving south-east to the plant frontage at 23.974° N. Between it and the shore lies a dark (deeper) **intake basin** that opens to the Gulf on the west/south-west. A ~250–300 m wide **dredged channel** runs east along the plant frontage, where **four stepped intake structures** (each ~300 m of frontage) line the shore at 52.226–52.238° E. At the east end, a visibly turbulent **outfall** discharges warm water into an **eastern channel** that runs north-east between breakwaters, well away from the intake — consistent with ENEC's stated separation design.




Fig. 1 — Intakes 1–4 on the traced geometry (website, Plate 05 step 4). Intake numbering is west → east and is ours; it is not ENEC's unit numbering.

> UNVERIFIED FEATURE — ASK ENEC
>
> Imagery also shows a ~3 km straight linear feature crossing the intake basin from about 52.200° E, 23.990° N to the shore near 52.212° E. It could be a boom, a net line, a pipeline or a cable. We do not depend on it, but if it is an existing jellyfish boom, High Tide can sit seaward of it and turn a passive barrier that clogs into an active one that sheds animals.

## The water

| PARAMETER | VALUE | NOTE |
|---|---|---|
| SST climatology at Barakah | 21.6 (winter) · 25.3 (spring) · 32.7 (summer) · 30.6 °C (autumn); daily P05/P50/P95 19.5/28.5/34.2 °C | GHRSST 2003–2019 (Nesterov 2021); trend +0.014 °C/yr |
| Salinity | 43.97 psu (2017), 45.01 psu (2019) | SCAD/EAD summaries; pH 8.04, DO ~4.9–5.1 mg/L |
| Mean current | ~0.05 m/s depth-averaged (Buoy 9, ~20 m depth) | Direction, maxima and approach velocities not given |
| Tidal currents | M2 4.28, K1 2.27, S2 2.10, S1 1.94, N2 1.12 cm/s | >85 % of tidal energy; tide-only reconstruction weak (r = 0.50) → non-tidal forcing matters |
| Model skill nearby | SCHISM RMSD 0.044 m/s, r = 0.352 | Regional context only |
| Winds | Shamal (north-westerly) episodes; sea breeze | ERA5 for history; local measurement needed |

## Life around the plant

- **Dugongs:** Abu Dhabi hosts the world's second-largest population (~3,000–3,500 in recent EAD surveys), concentrated in the Marawah Marine Biosphere Reserve (~100 km north-east of Barakah) and Al Yasat.
- **Turtles:** hawksbill and green turtles in Abu Dhabi waters; EAD reported a >30 % rise in sea-turtle numbers in its latest survey.
- **Whales and dolphins:** EAD recorded a Bryde's whale near Barakah; Indo-Pacific bottlenose and humpback dolphins occur regionally.

- **Barakah reef:** ENEC/Nawah surveys report nearly 100 species on the breakwaters and artificial reef, including hawksbill turtles; ENEC funds coral cultivation with EAD.

Why it matters: anything placed in the water must be wildlife-safe by design (no entanglement, low noise, no toxic exposure) and must not create new hard substrate for jellyfish polyps without a cleaning plan (chapter 17).

# The blue blubber — and a naming problem.

## Identity

ENEC's brief names ***Catostylus mosaicus*** (blue blubber, jelly blubber) with a 30–45 cm bell. A UAE University dissertation (Aravind et al., 2019–2022 sampling) reports *C. mosaicus* among Abu Dhabi taxa. However, the World Register of Marine Species lists ***C. perezi*** as a distinct species with an Arabian Gulf type locality, a 2024 integrative taxonomy paper illustrates *C. perezi* from Ras Al Khaimah, and Qatar University reported *C. perezi* as a first record for Qatar (2023). **Our position:** treat "Gulf *Catostylus*" as the design target, confirm local identity with vouchers and COI/16S barcoding (deliverable D6), and never let one species' response curve stand for a mixed bloom.

## Body plan

- **Order Rhizostomeae:** no marginal tentacles; eight fused oral arms with many tiny mouth openings; firm, dense mesoglea (≈95–97 % water).
- **Size:** brief target 30–45 cm bell. Australian *C. mosaicus* commonly 25–30 cm, up to 35+ cm.
- **Mass:** no published Gulf length–weight curve for this size. Indian *Catostylus* cf. *perezi* (n = 6,096): 2.6–29.8 cm, 4–1,822 g, mean 1.12 kg at 20.5 cm. Cube-scaling to 30–45 cm gives ≈1.9–12 kg; Portuguese *C. tagi* is typically ~2.5 kg at 25 cm. We use **4 kg (2–8 kg)** ASM.
- **Sting:** generally mild to humans; the hazard to the plant is mass, not venom.

## Senses and swimming

- **Rhopalia:** sensory structures on the bell margin carrying statocysts (gravity/balance) and, in many scyphozoans, simple light-sensitive ocelli. Jellyfish have a diffuse nerve net, no brain.
- **Chemosensing:** cnidarians detect dissolved molecules — e.g. reduced glutathione triggers the feeding reflex in *Hydra* (half-max 10 µM, Lenhoff 1961); moon and crystal jellies chose the Artemia side of a two-inflow tank (Arai 1991).
- **Swimming:** "pulse-and-glide" rowing. Neil & Askew (2018): five lab *C. mosaicus* with 2.3–4.6 cm bells, 74 bouts, mean 36.2 ± 1.2 mm/s at 2.18 Hz. Adult speeds are not measured; larger rhizostomes likely swim faster in absolute terms.
- **Active, not passive:** adult *Rhopilema nomadica* and *Rhizostoma pulmo* swam faster against a flume current than with it (Malul et al. 2019); field *R. nomadica* moved coherently against waves. Jellyfish are not just drifters — which is both our opportunity and a confound (rheotaxis must not be credited to chemistry).

## Life cycle — why blooms happen

Scyphozoans alternate between a **benthic polyp** (attached to hard surfaces) and the **free-swimming medusa**. Polyps bud asexually and, when triggered (often temperature change), **strobilate** — stacking and releasing many tiny ephyrae that grow into medusae over weeks. Blooms are therefore partly a "polyp-bank" phenomenon: lots of hard substrate (piers, breakwaters, pontoons), warm water, eutrophication and reduced predators/competitors (overfishing) can all raise bloom size. In Australia, *C. mosaicus* biomass in single estuaries ranges from ~1,800 to >18,000 tonnes (Pitt & Kingsford 2003).

## Seasonality in the UAE

Aravind et al. (2019–2022) report the taxon year-round, with aggregations in **October–November** on the Abu Dhabi coast (February–March in Ras Al Khaimah). Press reports describe blue blubbers in April–May elsewhere in the UAE. Temperature did not significantly predict bell size in their sample (Spearman, n = 130, $p > 0.05$). The often-quoted 20–35 °C "range" is not a steering threshold.

SPECIES-LEVEL FACTS WE SHOULD NOT OVERCLAIM

- Swim speed of a 40 cm adult — unknown.
- Whether Gulf *Catostylus* responds to nonenal — unknown.
- Local polyp locations — unknown (could include Barakah breakwaters; worth asking ENEC/EAD).
- Mass per animal at Barakah — unknown; ENEC's screen-wash records could give it.

# Soft animals, hard consequences.

Jellyfish are mostly water, but they deform and pack against screens, forming a near-impermeable mat. Flow falls, the pressure difference across screens rises, pumps lose suction margin, condenser vacuum degrades, and protection systems trip the unit. Cleaning is manual and the animals die.

## The chain of failure

1. Bloom arrives on tide and wind; the intake draws it in.
2. Coarse bar racks/trash racks catch large animals; travelling band or drum screens catch the rest.
3. Screens clog faster than spray-wash can clear them; differential pressure alarms.
4. Circulating-water flow drops; condenser back-pressure rises; operators derate or the unit trips automatically.
5. Crews remove tonnes by hand, vacuum trucks and divers; restart takes hours to days.

## Incident record (selected)

| YEAR | SITE | WHAT HAPPENED |
|---|---|---|
| 1999 | Philippines power plant | Jellyfish “enough to fill 50 trucks” drawn into the cooling system (POWER) |
| 2005, 2013 | Oskarshamn, Sweden | Unit 3 (1,400 MW) shut 29 Sept 2013 after tonnes of moon jellyfish clogged intake; ~2–3 days. Unit 1 in 2005 |
| 2006 | Chubu Electric, Japan | Intake shut automatically; output reduced |
| 2008, 2012 | Diablo Canyon, California | Oct 2008 ~1,000 moon jellyfish: Unit 2 offline, Unit 1 at half power. Apr 2012: salps forced Unit 2 shutdown. Plant uses rolling screens and can deploy an air bubble curtain |
| 2011 | Torness, Scotland | Both reactors shut ~1 week; ~£1 M/day lost. Again in 2021 |
| 2011 | St Lucie, Florida; Shimane, Japan; Orot Rabin, Israel | Shutdowns or derates from jellyfish (EPRI) |
| 2015 | Ashkelon (Rutenberg), Israel | Swarm cleared from screens with extra staff; drones used for monitoring |
| 2017 | Al-Zour, Kuwait | Jellyfish swarms at the power and desalination plant intakes (PAAET paper) |
| n.d. | Ghubra & Birka, Oman | 300 t of jellyfish damaged screens at Ghubra; 50 % output loss; Muscat water affected. Oman LNG estimated ~USD 7 M/day for a jellyfish stop |
| Aug 2025 | Gravelines, France | Units 2, 3, 4 tripped automatically when filter drums packed; Unit 6 next morning; 4 × 900 MW offline |
| Aug–Sept 2026 | Gravelines, France | Second swarm: Units 3, 4, 6 shut; 1, 2 reduced. EDF runs 7 boats round the clock with France Pêche Durable to collect and release swarms offshore |

WHAT WE KNOW ABOUT BARAKAH

We are not aware of a publicly reported Barakah unit trip caused by jellyfish. ENEC has a detection programme and cited ~200 t/day historic loads in mentoring, which implies the current system copes — at the cost of manual effort and animal mortality. Our pitch is **insurance and less manual work**, not “Barakah is failing”.

# The defence landscape.

| APPROACH | HOW IT WORKS | EVIDENCE / USERS | STRENGTHS | WEAKNESSES |
|---|---|---|---|---|
| **Screens, racks, drums** | Physical filtration at the intake | Every plant; Barakah has a refined design | Certain, last line | Clog under mass loads; kills animals; manual cleaning; trips (Gravelines) |
| **Nets / booms** | Barrier across intake bay or breakwater heads | Gulf desal & power; Elkamash & Baawain (2010) proposed screens at breakwater heads | Cheap, passive | Clog and block water; maintenance boats; kills animals; storms |
| **Boats collecting** | Trawl/scoop and release offshore | EDF Gravelines: 7 boats, 24/7 (2026) | Animals released alive; flexible | Manual, night/weather limits, crew cost, scale |
| **Shredder robots** | Autonomous surface robots net and grind | KAIST JEROS: 3 robots ≈ 0.9 t/h (2013) | Autonomous | Kills; critics note shredding mature medusae can release eggs and larvae; low throughput: ~28 robots for Barakah's mean load |
| **Fixed bubble curtain** | Seabed hose; upwelling and surface current deflect drifting animals | Diablo Canyon, Ringhals; Mowi Scotland salmon farm (≈80 % fewer jellies inside); Bubble Tubing® since 2007 | No contact, harmless, proven hardware | Always on (energy), one direction; Lo (1991) lab: barrier alone inadequate — net recommended behind it |
| **Water jets / crossflow** | Pumped jets push surface water sideways | Kuwait PAAET concept (drones, LiDAR, water jets) | Directional | Pumping power; entrains animals |
| **Intake geometry** | Lower approach velocity so weak swimmers drift past | EDF EPR2 design via CFD (planned) | Permanent | Only for new builds |
| **Early warning** | Satellite, drones, sonar, eDNA, models | ENEC programme; Israel Electric drones | Time to prepare | Does not move animals — the gap ENEC set us |
| **Chemical deterrents** | Repel/attract by smell | **None deployed at power plants that we know of** | Acts on behaviour, low energy | Unproven for adults; environmental review — **our differentiator and our biggest risk** |

## Where High Tide is genuinely new

1. **Controlled, segmented, tide-aware routing** instead of an always-on, one-direction curtain.
2. **Verification**: we count clears and returns, not just "deflected".
3. **Behavioural cues on top of physics**, tested in a 2 × 2 design that isolates their added value.
4. **Siting by the "pull vs swimmer" rule**: put the gate where animals can still win.

The individual parts (bubble curtains, the molecule, Artemia attraction) come from prior work. The contribution is linking them into one controlled, verified system and measuring whether chemistry adds enough to justify itself.

08 · HIGH TIDE IN ONE PICTURE

# Three prongs and a controller.

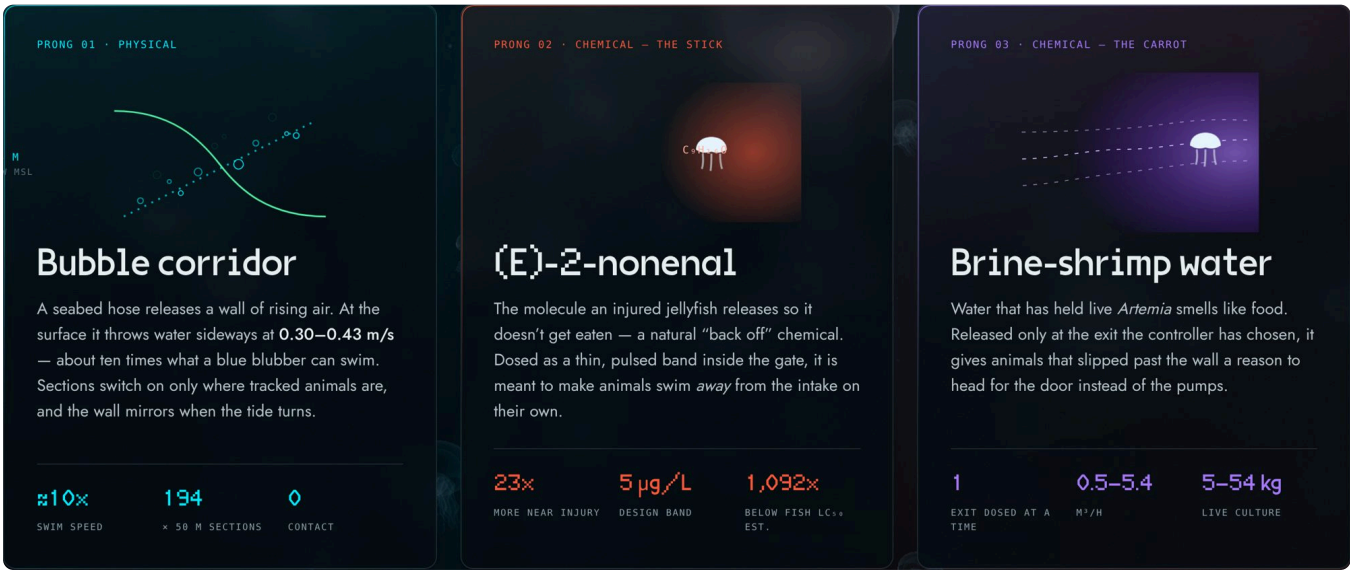



Fig. 2 — The three prongs as presented on the website.

## Concept of operations (one event)

| TIME | WHAT HAPPENS | WHO / WHAT |
|---|---|---|
| T – hours | ENEC offshore alert received; controller arms; compressors pre-start; ADCPs report current direction; cue stock checked | Controller, operator acknowledges |
| T – 30 min | Camera masts and gate sonar acquire first tracks at the basin mouth | Sensing |
| T | Tracks within 13 m of the gate: their 50 m section and neighbours fire; route chosen from tide (flood → line A / Exit N; ebb → line B / Exit W) | Controller |
| T + … | Carrot runs at the active exit; stick band pulses behind the gate only when tracks are inside the gate | Dosing skid (after gates G3–G4 only) |
| Tide turn | Controller waits 10 s after the measured reversal, switches lines, swaps carrot station | Controller |
| Breach | Tracks past the gate approaching the channel: guard double-curtain fires | Controller |
| Always | Counts: arrivals, cleared, retained, returned, intake entries; logs air, power, dosing, water quality | Historian |
| All-clear | No tracks for N minutes; air and dosing off; report to ENEC | Operator |

## Fail-safe philosophy

Every failure mode drives the system to **off**: air off, valves closed, dosing closed (fail-closed source, no-discharge waste path). The plant's existing screens are untouched and remain the protection of record. High Tide reduces the load on them; it never replaces them. Nothing is installed inside the intake structures.

# A wall made of air.

## How a bubble curtain moves water

Compressed air leaves a perforated seabed hose as a line of bubbles. The rising bubbles drag water up with them (a line plume). At the surface the upwelling splits and flows away from the line on both sides as a **horizontal surface current**, a few metres thick, with return flow underneath. Drifting animals near the surface meet an outward current that holds them back; with any along-line component of the background flow, they slide along the line to its downstream end. That is deflection. No contact is needed.

```
Bulson (1961):   V_m = 1.46 · (g · q)^(1/3) · (1 + D / H₀)^(1/3)
  V_m  surface current at ~one depth from the line (m/s)
  q    free-air flow per metre of hose (m²/s)     D  water depth (m)     H₀ ≈ 10.3 m (1 atm of water)
```

| AIR PER METRE (BEELEN ET AL. BASIN RANGE) | DEPTH 6 M | DEPTH 8 M |
|---|---|---|
| 0.55 L/s·m (low) | 0.30 m/s | 0.31 m/s |
| 1.11 L/s·m (design midpoint) | 0.38 m/s | 0.39 m/s |
| 1.67 L/s·m (high) | 0.43 m/s | 0.45 m/s |

Ratios that matter: surface current ÷ swim speed ≈ **10×**; surface current ÷ mean intake pull at the gate (0.03 m/s) ≈ **13×**; in the channel (0.27–0.29 m/s) the ratio collapses to ~1.3×, which is why the guard line uses ~3.8 L/s·m for a 0.6 m/s jet and why the gate sits at the mouth.

## Evidence that it works on jellyfish

- **C-CORE / Thodi et al.:** large-scale tests with silicone jellyfish (5 and 18 cm), currents 0.1–0.6 m/s, curtain angles 5°/30°/45°, air 0.1–0.6 cfm/ft (≈0.16–0.93 L/s·m), 1–3 curtains. Effective deflection achievable up to 0.6 m/s; angle, number of curtains, spacing and air flow all helped at high current; CFD matched tests.
- **Aquaculture:** Mowi's Portnalong (Scotland) site — jellyfish/plankton inside the curtain up to 80 % lower than outside on many days.
- **Power plants:** Diablo Canyon can deploy an air bubble curtain in its intake bay; Ringhals (Sweden) has one (Cal Poly anchoring study).
- **The caution:** Lo (1991) lab work concluded a bubble barrier alone was not adequate protection and recommended a net behind it. Our answer: keep the screens as the backstop, add a guard line, route rather than block, and add cues. EPRI field tests on *fish* (Roseton 1986–87; Arkansas Nuclear One 1974–75) found bubble curtains ineffective for fish — fish are strong swimmers that habituate; drifting jellyfish are the opposite case.

## Design choices and why

| CHOICE | VALUE | WHY |
|---|---|---|
| Angle | 15–30° off square to the flow (model uses 30°) | Thodi: angled curtains deflect better at high current; angle gives an along-line slide toward the exit |
| Two mirrored lines | Line A (flood → Exit N), Line B (ebb → Exit W) | The exit must be downstream of the tidal drift; a single fixed line piles animals up when the tide turns (model: 722 retained, 7,875 return events) |
| Section length | 50 m | Matches camera/sonar tracking resolution; 179 gate sections; keeps valve count manageable |
| Trigger | Track within 13 m of the line → its section ± neighbours | From v1.0 controller; saves air when the bloom is patchy |
| Switch delay | 10 s after measured reversal | Avoids chattering at slack water; 90 s delay tested (intake entries 78 → 92, seed 42) |
| Air | 1.11 L/s·m at 6 m (gate); 3.78 L/s·m at 9 m (guard) | Beelen midpoint; guard sized by Bulson for 0.6 m/s |
| Hose | Weighted perforated HDPE / EPDM slit-sleeve diffuser | Slit sleeves close when air stops (less clogging); HDPE resists corrosion |

| | | |
|---|---|---|
| Anchoring | 60–100 kg clumps every 5 m; weak-link tethers | Recoverable; no hose can become intake debris |
| Air quality | Oil-free compressors | No oil sheen; required for marine release |

## Gulf-specific engineering

- **Heat:** 45–50 °C summer ambient derates compressors and raises discharge temperature; aftercoolers and shaded, ventilated enclosures; size at 45 °C.
- **Biofouling and scaling:** warm, saline Gulf water fouls fast; periodic purge cycles, a trickle flow to keep slits open, annual diver inspection, replaceable sleeves. (Bonus: Miura et al. found the same family of molecules inhibits bacterial biofilm — not a design claim.)
- **Waves and shamal:** waves break up the surface jet; the controller logs wave state and the plant falls back to screens in storms (an operating-envelope boundary to agree with ENEC).
- **Salinity 44–45 PSU:** denser water slightly changes bubble rise and plume entrainment; bench work runs at 44–46 psu.
- **Dissolved oxygen:** curtains aerate (DO goes up slightly) — neutral to positive locally.
- **Sediment:** plumes can resuspend fine sediment near the hose; place on firm bed, check turbidity at the gate sonde.

## Pilot sizing (already in the dossier)

| CASE (10 M AT 8 M DEPTH) | FREE AIR | COMPRESSOR INPUT |
|---|---|---|
| Low 0.55 L/s·m | 0.33 $m^3$/min | 0.71 kW |
| Mid 1.11 L/s·m | 0.67 $m^3$/min | 1.43 kW |
| High 1.67 L/s·m | 1.00 $m^3$/min | 2.15 kW |

Compression: adiabatic, inlet 101.3 kPa, outlet = atmospheric + ρgD + 20 kPa line/diffuser losses, 60 % combined efficiency. Verify on supplier curves.

# (E)-2-nonenal.

## Identity

| PROPERTY | VALUE |
|---|---|
| Name | (E)-non-2-enal, trans-2-nonenal, "cucumber aldehyde" |
| Formula / mass | $C_9H_{16}O$, 140.22 g/mol |
| Identifiers | CAS 18829-56-6 · PubChem CID 5283335 · InChIKey BSAIUMLZVGUGKX-BQYQJAHWSA-N · SMILES CCCCCC/C=C/C=O · FEMA 3213 · JECFA 1362 |
| Physical | Colourless–pale yellow liquid; density 0.846 g/mL; bp ~188–190 °C; flash point ~84 °C; air-sensitive (oxidises) |
| Water behaviour | Est. solubility ~205 mg/L at 25 °C; log P ≈ 3.1–3.3 (moderately hydrophobic); vapour pressure ~0.02–0.26 mmHg (volatile) |
| Uses | Flavour and fragrance (cucumber, melon); suggested food use 0.1–10 ppm; also the "old-person odour" aldehyde in skin-lipid oxidation |
| Occurrence | Cucumbers, melons, oxidised fats; brown algae and dinoflagellate cultures (background in seawater possible) |
| Hazard | Skin/eye irritant (SDS). One ingredient SDS lists aquatic chronic category 2 with ECOSAR estimates: fish $LC_{50}$ ≈ 5.5 mg/L, algal 96 h $EC_{50}$ ≈ 8.0 mg/L |

Do not confuse with 4-hydroxy-2-nonenal (a toxic lipid-peroxidation product), nonanal, or (E,Z)-2,6-nonadienal. Their safety data are not interchangeable.

## The evidence, precisely

**Miura et al. 2026, *Journal of Chemical Ecology* 52:64** (Tokyo University of Marine Science and Technology with Enoshima Aquarium). Prey: the hydromedusa *Spirocodon saltatrix*; predator: the Japanese sea nettle *Chrysaora pacifica*.

- *S. saltatrix* escapes by an "involution" behaviour; removing its exumbrella skin made it edible (14 of 15 de-skinned eaten vs 0 of 15 sham, $p = 1.1 \times 10^{-7}$) → the skin is defensive.
- SPME–GC–MS of water around **injured** vs **sham** skin: (E)-2-nonenal **23×** higher (n = 10/group, **p = 0.00094**), 1-nonanol 2.2×, (E)-2-nonen-1-ol 6.6×.
- Feeding deterrence: krill treated with nonenal at 50 % and 5 % formulations were released faster by the sea nettle; 0.5 % not significant. (E)-2-octen-1-ol and 1-octen-3-ol worked down to 0.5 %.
- Biofilm: several compounds inhibited a bacterial biofilm >90 % at 5 %.
- The authors note measured bulk concentrations were below the effective concentrations; they hypothesise micro-scale concentrations at the wound reach the threshold.

> HOW TO EXPLAIN THE STATISTICS TO A NON-SCIENTIST
>
> **"23×"** — the water around an injured jellyfish had 23 times more of this molecule than the water around an unhurt one.
> **"n = 10 per group"** — ten injured samples and ten unhurt samples were measured.
> **"p = 0.00094"** — if there were really no difference, a gap this large would appear by chance less than 1 time in 1,000.
> **What it does NOT say** — that any jellyfish swims away from it, or what concentration in open water does anything. It is a deterrent at contact, in a different species.

## From evidence to hypothesis

Our hypothesis: a localised gradient of nonenal acts as a **directional** repellent for adult Gulf *Catostylus*. Four separate claims must not be blurred: (1) chemical sensing, (2) changed pulsing, (3) a short-term turn, (4) safe sustained diversion. Only (4) matters to ENEC. Activity suppression, stunning, aggregation or injury count **against** the candidate.

Terminology to use precisely: a **pheromone** is within-species; a **kairomone** benefits the receiver across species (e.g. smelling a predator); an **allomone** benefits the emitter (a defence chemical such as this one); a **feeding deterrent** acts after contact.

## Dose design envelope ASM

```
mass rate = c_target × u × L_active × h × duty
  c_target  band concentration (µg/L = mg/m³)      u  water speed through the band
  L_active  active band length (300 m of 600 m)     h  surface layer (3 m)       duty  40 % pulsing
```

| BAND TARGET | WATER SPEED | G/H | KG PER 12 H | AT PUMPS MG/L | BAND VS FISH $LC_{50}$ |
|---|---|---|---|---|---|
| 1 µg/L | 0.05 / 0.1 / 0.2 m/s | 65 / 130 / 259 | 0.8 / 1.6 / 3.1 | 0.03 / 0.05 / 0.10 | 5,461× |
| 5 µg/L | 0.05 / **0.1** / 0.2 m/s | 324 / **648** / 1,296 | 3.9 / **7.8** / 15.6 | 0.13 / **0.26** / 0.52 | 1,092× |
| 10 µg/L | 0.05 / 0.1 / 0.2 m/s | 648 / 1,296 / 2,592 | 7.8 / 15.6 / 31.1 | 0.26 / 0.52 / 1.04 | 546× |

"At pumps" assumes the dosed water is fully mixed into the 690 $m^3/s$ intake flow — a worst case for the plant, because everything ends up there. 5 µg/L ≈ a teaspoon in a 25 m pool. Carrier: pre-dilute to 100 mg/L (half the estimated solubility) in filtered seawater → ~16 $m^3/h$ while dosing.

## Fate in the water

- **Volatilisation:** moderately volatile; surface release loses mass to air (reduces exposure; also reduces efficacy).
- **Oxidation:** α,β-unsaturated aldehyde; oxidises to 2-nonenoic acid; air-sensitive in storage.
- **Sorption:** log P ~3.2 → some partitioning to organic matter and sediment; low bioaccumulation expected but must be measured.
- **Through the plant:** water passes the condensers (+~7–10 °C typical ΔT) and leaves at the outfall; heat accelerates loss. Concentrations at the outfall would be below the 0.26 µg/L intake mix.

## Safety and regulation

- No chemical goes into Barakah waters until gates G3 (efficacy) and G4 (non-target safety) pass, ENEC and EAD agree thresholds, and a contained trial proves capture.
- Non-target screen: marine microalgae growth (ISO 10253), copepod mobility/survival, bivalve larval development (ISO 17244), a local non-target cnidarian, a fish endpoint chosen by specialists. Pass = no effect at **10× the design band**.
- Selectivity ratio = lowest credible non-target effect ÷ effective steering dose. Undefined today; the assay programme measures it.
- EAD regulates marine discharges via environmental assessment and licensing; any novel release needs explicit approval. FANR regulates nuclear safety; we install nothing in the intake structures.

## Second candidate and rejects

**(E)-2-octen-1-ol** was active at 0.5 % in the feeding assay but was *not* detected in injured-water samples — keep both facts visible. **Rejected:** ammonium chloride (attracted *Aurelia* historically, but ammonia toxicity in warm, high-pH water makes it unacceptable near an intake); indoles that trigger strobilation (would make more jellyfish); live predators, extracts, antibiotics, general toxicants.

# Brine-shrimp water.

## The evidence

**Arai (1991)**: in a two-inflow, central-outflow aquarium, *Aurelia aurita* and *Aequorea victoria* medusae favoured the side containing screened *Artemia*; *Aurelia* also favoured Artemia-conditioned water (and ammonium chloride) in one inflow. Flows were balanced and sides swapped to control for position bias. This is the direct precedent for a chemical side preference in medusae.

Related: Gimenez, Carroll & Pitt (2026) showed *Aurelia coerulea* planula settlement fell in water carrying predator (nudibranch) cues — chemical sensing at another life stage. A 2026 two-choice study found *Aurelia* moved away from a conspecific behind mesh (chemistry not isolated).

## Design

- Standing culture of live *Artemia* on shore; seawater recirculates through it and is filtered (shrimp stay; only their "smell" leaves).
- Released at **one exit at a time** — the one the controller is routing to — via a small diffuser at Exit N (just outside the breakwater head) or Exit W (along the shore).
- Purpose: animals that slip past the wall or idle in the basin get a reason to head for the door. Attraction is only useful if the exit truly clears — otherwise you build an aggregation.

## Quantities ASM

| DILUTION AT EXIT WINDOW | STOCK RELEASE | STANDING ARTEMIA | TANK |
|---|---|---|---|
| 1 : 100,000 (lean) | 0.15 L/s = 0.54 m³/h | 5.4 kg | ~1 m³ |
| 1 : 10,000 (strong) | 1.5 L/s = 5.4 m³/h | 54 kg | ~8 m³ |

Exit window: 50 m wide × 3 m deep, water at 0.1 m/s = 15 m³/s. Stock: 10 g wet Artemia per litre per hour of conditioning. The bench series (0 / 10 / 50 / 100 % stock blended into lab feed) finds the real threshold.

## Risks specific to the carrot

- **Attracting the wrong animals** (fish, other jellyfish) — modest volume, filtered; monitor at the exit sonde and camera.
- **Nutrient load** — dissolved organics and ammonium from culture water; at 0.5–5 m³/h this is tiny next to 690 m³/s, but it is logged.
- **Biosecurity** — Artemia cysts from certified suppliers; culture water filtered; no live shrimp released.
- **Aggregation at the exit** — counted as "retained/returned" and is a failure signal; carrot shuts off if returns rise.

# Put the gate where the animal can still win.




Fig. 3 — The point of no return: mean intake pull equals swim speed ~4.3 km from the intakes. White dots: bloom; red: animals on a one-way trip.

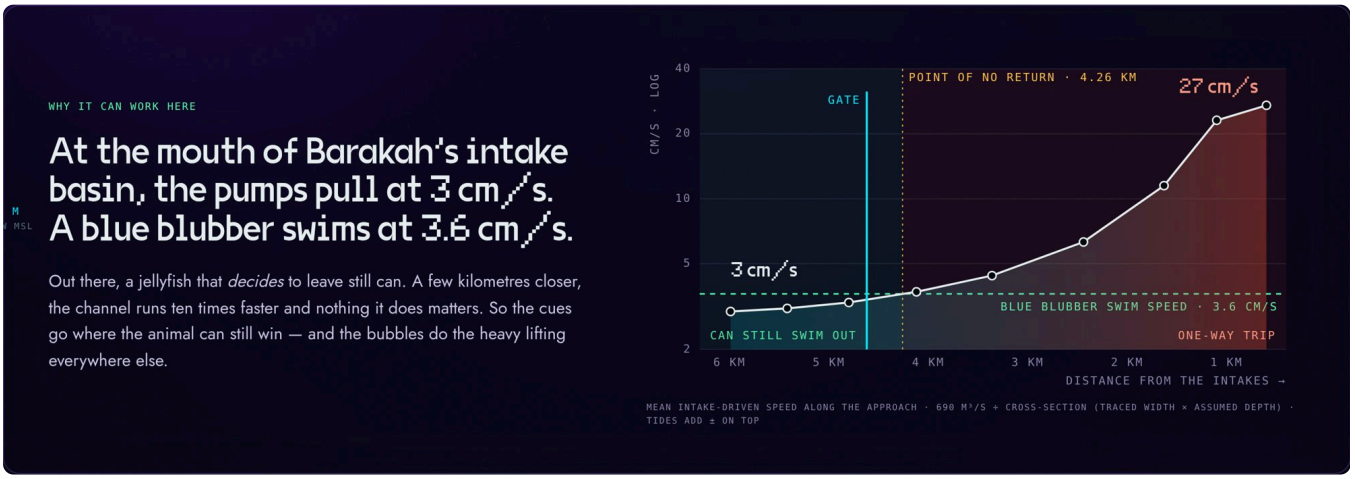



Fig. 4 — Mean intake-driven speed along the approach (log scale). 690 m³/s ÷ (traced width × assumed depth 5.5–8.5 m). Tides add oscillating ± on top.

| KM FROM INTAKES | WIDTH (TRACED) | DEPTH (ASM) | MEAN PULL |
|---|---|---|---|
| 5.99 | 4,162 m | 5.5 m | 3.0 cm/s |
| 4.80 | 3,797 m | 5.5 m | 3.3 cm/s |
| 4.12 | 3,230 m | 5.8 m | 3.7 cm/s |
| 3.36 | 2,512 m | 6.3 m | 4.4 cm/s |
| 2.44 | 1,598 m | 6.9 m | 6.3 cm/s |
| 1.63 | 807 m | 7.4 m | 11.5 cm/s |

| | | | | |
|---|---|---|---|---|
| | 1.10 | 384 m | 7.8 m | 23.1 cm/s |
| | 0.60 | 300 m | 8.5 m | 27.1 cm/s |

Drift time from the gate to the screens on the intake-driven mean flow alone: **≈19 h** [MOD]. Tides slosh water back and forth on top of that, so hour-to-hour motion is tidal; the mean says there is time for the controller, cues and guard to act.




Fig. 5 — Full concept layout (website, Plate 05 step 7). Positions are indicative; to be sited with an ENEC survey.

## Layout elements

| ELEMENT | WHERE | SIZE | ROLE |
|---|---|---|---|
| Gate line A | From the shore at ~52.180° E (Exit W apex) to the west breakwater head (~52.191° E, 24.002° N) | 4,366 m · 88 sections | Flood tide: drift slides NNE along A, round the head (Exit N) |
| Gate line B | From the west breakwater at ~52.202° E to the same shore apex | 4,542 m · 91 sections | Ebb tide: drift slides WSW along B to the shore apex (Exit W) |
| Exit N | Outside the breakwater head | — | Flood current carries animals ENE along the outside of the breakwater, away from the basin |
| Exit W | Along the shore, west of the mouth | — | Ebb current carries animals WSW along the coast |
| Stick band | ~700 m inside the gate, on the basin axis | 600 m, pulsed | Pushes animals that crossed the gate back out |
| Carrot N / W | At each exit | One active at a time | Pulls animals toward the open exit |
| Guard | Across the channel entrance in front of Intake 1 | 2 × 375 m, 15 sections | Fires on breach only |
| Camera masts | Breakwater head; breakwater bend | 2 (EO/IR) | Track bells by day |
| Imaging sonar | Gate | 2 | Night / turbid tracking |

| | | | |
|---|---|---|---|
| ADCPs | Gate, basin, channel | 3 | Current direction for routing |
| Shore station | West of the plant compound | 5 × 250 kW, dosing skid, culture | Air, dosing, control; feeders routed on seabed and breakwater crest |

## Alternatives we considered

- **Gate at the channel throat** (~0.8 km wide): short, cheap — but exits are dead ends at the breakwater and the pull there is 4–12 cm/s, faster than the animal swims. Rejected as primary; kept as the guard.
- **Gate inside the basin (~2.6 km span)**: north end hits the breakwater wall; would need a ~3.6 km "bubble conveyor" of angled segments along the breakwater to walk animals out. Similar hose length to the mouth gate with a weaker exit.
- **Single V (chevron) gate**: half the hose, both exits always open; not modelled yet — a good cost-down option to test (Q&A).
- **Phase-1 lean build**: one line (B, ebb) + guard ≈ 5.3 km of hose — roughly half CAPEX — if the tidal record shows one dominant arrival phase.

# High Tide v1.0 — what it is and isn't.

A deterministic, depth-averaged particle transport "thought experiment" that compares diversion strategies under matched conditions. Not CFD. No biological calibration. No crowding. Its job is to rank strategies and reveal failure modes, not to predict tonnes.

## Domain and physics

| ITEM | VALUE |
|---|---|
| Domain | Approach strip 60 m wide (y = 0–60), arrivals at x = 0; plume line through (43, 30); intake at x ≥ 95 m, 18 ≤ y ≤ 42; exits at y < 0 or y > 60; other boundaries x > 140, x < −35, y < −60, y > 120 |
| Time | 900 s, Δt = 0.25 s; 1 arrival per second (×3 in second half for surge) |
| Water | Along-flow 0.22 m/s; cross-flow ±0.025 m/s (sign flips at 450 s in reversal); intake pull 0.22 m/s × Gaussian in y (σ 18 m) × ramp beyond x = 55 m |
| Dispersion | Deterministic sinusoidal "unresolved motion", 0.035 m/s (0.15 in high-dispersion) |
| Curtain | Six 10 m sections on a 30° line; symmetric surface spreading 0.55 m/s × response × Gaussian (σ 5 m) about the line |
| Animal response | Response factor 0.7–1.3 per animal, scaled by (0.375 / bell)^0.25; zero-response scenario sets it to 0 |
| Controller | Track within 13 m of the line → its section and neighbours; route sign switches 10 s (or 90 s) after reversal; off on fault or closed exit |
| Cue layer (separate module) | Carrot: lateral pull toward the routed exit; stick: Gaussian plume in front of the intake pushing out; combined bias capped at 36.2 mm/s × response |
| Configurations | none · fixed · controlled · angled porous guide (15 % passage) · distributed water flow |
| Outputs | intake, outside at finish, retained, other boundary, ever exited, returned, return events, contact events, actuation index, balance error (always 0) |




Fig. 6 — Reversal run, seed 42 (cue row: same seed, cue layer on).

## Results across scenarios (3-seed means, of 900 arrivals unless noted)

| SCENARIO | NONE: INTAKE | CONTROLLED: INTAKE | +CUES: INTAKE | CONTROLLED: CLEARED | +CUES: CLEARED |
|---|---|---|---|---|---|
| Steady approach | 296 | 0 | 0 | 389 | 392 |
| Reversal at 450 s | 339 | 86 | 12 | 253 | 301 |

| | | | | | |
|---|---|---|---|---|---|
| Reversal + 90 s delay | 339 | 97 | 14 | 198 | 235 |
| Surge ×3 (1,800 arrivals) | 379 | 0 | 0 | 518 | 638 |
| Weak flow | 0 | 0 | 0 | 32 | 234 |
| Strong intake pull | 339 | 0 | 0 | 370 | 385 |
| High dispersion | 310 | 0 | 0 | 388 | 398 |
| Power loss at 450 s | 296 | 176 | 156 | 6 | 11 |
| Return current | 339 | 86 | 12 | 66 | 68 |
| Blocked exit | 296 | 296 | 296 | 0 | 0 |
| Zero response | 296 | 296 | 296 | 6 | 6 |




Fig. 7 — What the cue layer adds (hypothesis): reversal intake entries, weak-flow clearance, weak-flow air use.

## How to read the results honestly

- **"Fixed bubbles: 0 intake" in reversal is not a win.** 722 animals stay trapped and bounce back 7,875 times — a pile-up waiting for the next tide. Controlled sections trade 78 entries for 266 actually leaving.
- **Blocked exit and zero response give 0 % improvement** — by design. They prove the model is not rigged and show what to verify on site (a real exit, a real response).
- **Power loss:** the system fails off and returns to the no-diversion baseline from that moment — the screens carry the load.
- **Return current:** clearing is fragile if water returns from outside — instrument returns.

## How the 60 m model relates to the 4 km gate

The model is dimensional but generic: a 60 m strip with a 0.22 m/s approach flow and a 0.55 m/s plume (ratio 2.5). At the real gate, our estimates give a 0.03 m/s mean approach and 0.38 m/s bubble current (ratio ~13). So the model is **harsher** than the mouth location. Arrival density: the model's 1 animal/s into 60 m compares with ~1.7 animals/s at peak across a 4.2 km gate — the model's per-metre density is far higher. Treat the full gate as ~70 model strips side by side, each running the same logic.

> WHAT THE LIVE SIMULATION ON THE WEBSITE SHOWS
>
> The 3D scene is the v1.0 engine running unchanged in the browser (with the cue layer optional). With the cue layer off it reproduces the archived seed-42 results exactly. Staging (offshore approach, buoys, mast, domes) is illustrative. Scenario selector, speed and corridor modes are live.

# From 200 tonnes to things you can size.

| STEP | CALCULATION | RESULT |
|---|---|---|
| Peak day | ENEC mentoring figure | 200 t |
| Animals | 200,000 kg ÷ 4 kg (2–8 kg) | 50,000 (25k–100k) |
| Mean rate | 200 t ÷ 24 h | 8.33 t/h · 2,083 animals/h |
| Peak hour | × 3 (two ~6 h tidal windows × 1.5 intra-window) | 25 t/h · 6,250 animals/h · 1.74/s |
| Per intake | ÷ 4 (even split assumed) | 50 t · 12,500 animals |
| Water that day | 690 m³/s × 86,400 s | 59.6 million m³ |
| Mean concentration | 200 t ÷ 59.6 M m³ | 3.4 g/m³ · 1 animal per 1,192 m³ |
| Truckloads | ÷ 10 t | 20 |
| JEROS equivalent | 8.33 t/h ÷ 0.3 t/h per robot | 28 (83 at peak) |

## Sensitivity

- Animal mass is the biggest unknown: halving it doubles the animal count but not the tonnage. Hardware is sized by area and current, not by count; tracking load scales with count.
- Peak factor: if arrivals are concentrated in one 4 h window, peak = 50 t/h; the gate's capacity is not count-limited (no contact), but sensors must handle ~3.5 animals/s.
- Uneven intakes: the westernmost intake (first on the channel) probably sees more; ENEC screen data per unit would show it.

## Where 200 t goes with High Tide MOD

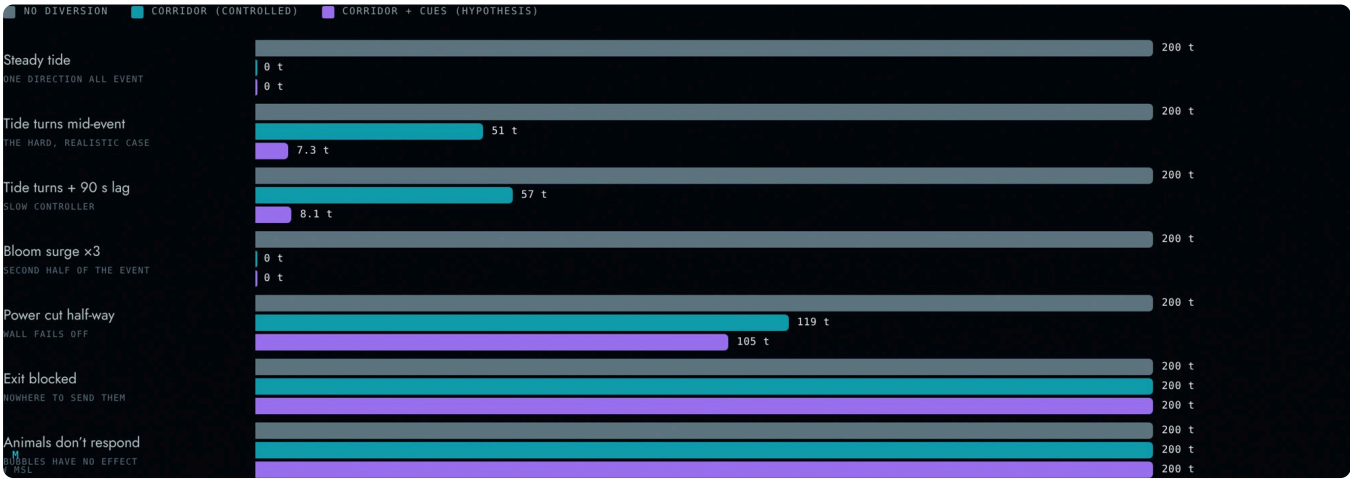



Fig. 8 — Tonnes reaching the screens if each run's intake fraction applied to a 200 t day.

# What gets built.

| LINE | LENGTH | DEPTH | AIR PER M | FREE AIR | POWER IF ALL ON | SECTIONS |
|---|---|---|---|---|---|---|
| Gate A | 4,366 m | 6 m | 1.11 L/s | 291 m³/min | 520 kW | 88 |
| Gate B | 4,542 m | 6 m | 1.11 L/s | 303 m³/min | 541 kW | 91 |
| Guard (double) | 749 m | 9 m | 3.78 L/s | 170 m³/min | 393 kW | 15 |
| **Total** | **9,657 m** | | | | | **194** |

| ITEM | QTY | NOTE |
|---|---|---|
| Diffuser hose / perforated HDPE (EPDM slit sleeves) | 9,657 m | All bubble lines incl. double guard |
| Ballast clump anchors (60–100 kg, every 5 m) | 1,932 | Weak-link tethers, recoverable |
| Section valves (pneumatic, shore valve house) | 194 | 50 m sections |
| Air feeder bundles HDPE DN50–DN200 | 20,359 m | Trunk + section cores; along seabed and breakwater crest |
| Oil-free screw compressors | 5 × 250 kW (N+1) | 1,250 kW installed |
| Air receivers, dryers, aftercoolers | 2 sets | 45 °C ambient design |
| ADCP current profilers | 3 | Gate, basin, channel |
| Camera masts (EO/IR) | 2 | Breakwater head + bend |
| Imaging sonar | 2 | Gate |
| Water-quality sondes (T, S, DO, pH, turbidity) | 3 | Gate, Exit N, intake |
| Stick dosing skid (tank, metering pump, mixer) + 600 m diffuser | 1 | ~16 m³/h carrier while dosing |
| Carrot culture + conditioning tanks | ~8 m³ | 2 release points |
| Controller (PLC + tracking PC + historian) | 1 + hot spare | Fail-safe: air off, dosing off |

## Energy

| CASE | KW | NOTE |
|---|---|---|
| Worst case: one full gate line + guard | 913 | 0.016 % of 5,600 MW |
| Typical event (85 % of sections active) | 442 | Model actuation index |
| Weak flow, corridor only (36 %) | 187 | |
| Weak flow, with cues (18 %) | 94 | Cue hypothesis |
| Per 12 h event | 5.8 MWh | ≈ AED 1,700 at AED 0.30/kWh (internal value ASM) |

# Insurance, not a cost-saving fantasy.

| ITEM | VALUE | BASIS |
|---|---|---|
| CAPEX full scale (Class 5: −50 % / +100 %) | **AED 18–67 M** | Hose AED 1,500–4,500/m installed (freshwater benchmark ≈ USD 1,250/m, Zielinski & Sorensen 2016, × marine factor); compressors AED 1,800–3,500/kW; sensors AED 0.9–2.2 M; dosing AED 0.15–0.45 M; controls AED 0.3–0.9 M; +30 % on high side |
| Value of one unit-day | **33,600 MWh ≈ USD 1.7–2.7 M ≈ AED 6.2–9.9 M** | 1,400 MW × 24 h × USD 50–80/MWh (ASM; Barakah PPA price not public) |
| Payback | **2–11 avoided unit-days** | CAPEX ÷ unit-day value, best/worst pairing |
| Event OPEX | Energy ~AED 1,700; stick AED 8,600–28,800 (once approved); carrot culture small | Bulk flavour-grade nonenal USD 300–1,000/kg (ASM) |
| Benchmarks | Torness ~£1 M/day (2011); Oman LNG ~USD 7 M/day (estimate); EDF 7 boats 24/7 | Press |

The commercial comparator ENEC should use: **cost per confirmed wet tonne diverted safely**, with the denominator, verification interval and failure cases defined — plus avoided crew-hours, disposal, screen wear and derate risk. Total cost per event = mobilisation + crew hours × loaded rate + energy × tariff + handling/disposal + apportioned maintenance + consumables. ENEC's own numbers would replace our assumptions.

## Pilot ladder

| STAGE | LOW (AED) | HIGH (AED) | DELIVERABLE |
|---|---|---|---|
| Bench system | 2,160 | 8,640 | Repeatable flow, section control, track logging |
| Contained biological feasibility | 35,625 | 125,000 | Directional response, recovery, initial non-target screen |
| Bench + contained | 37,785 | 133,640 | Paired physical baseline and measured cue result |
| One recoverable physical marine trial | 105,950 | 486,200 | Survey, monitored trial, report with clearance, returns, power |

Marine-trial allowance incl. 30 % reserve: survey & engineering AED 20–81k, instrument rental 6–48k, hardware & spares 30–100k, marine logistics & retrieval 10.5–90k, environmental observation 10–35k, data analysis 5–20k.

# Do no harm, and prove it.

## Effects of the bubble corridor

| RECEPTOR | POSSIBLE EFFECT | MITIGATION / EVIDENCE |
|---|---|---|
| Jellyfish | None by contact; possible bubbles under the bell | Recovery and condition scored in trials; animals routed alive |
| Fish | Mild deterrence/habituation (most fish learn to pass) | Literature on fish curtains; camera monitoring |
| Dugongs, turtles, dolphins | Low broadband noise; no entanglement (no nets) | Bubble curtains are used to *reduce* pile-driving noise; hose on seabed; weak-link anchors; marine-mammal watch protocol in trials |
| Water quality | Slight DO increase; local turbidity | Sondes at gate, exit, intake |
| Seabed / seagrass / coral | Anchor footprint; resuspension | Pre-installation habitat survey; route around sensitive patches |
| Jellyfish polyps | New hard surfaces could host polyps | Smooth HDPE, periodic inspection/cleaning; check existing breakwaters with EAD |

## Effects of the cues

Covered in chapters 10–11. Principle: **contained first**, measured concentrations, non-target endpoints at 10× design band, fail-closed dosing, no-discharge waste path in the lab, and EAD/ENEC agreement before any release. The first marine trial is physical only.

## Regulators and permissions

| BODY | RELEVANCE |
|---|---|
| ENEC / ENEC Operations (formerly Nawah) | Site owner/operator; access, safety cases, security clearance, operating envelope, data |
| FANR (Federal Authority for Nuclear Regulation) | Nuclear safety. We install nothing in intake structures; any device in the approach must be shown unable to become intake debris or affect the ultimate heat sink |
| EAD (Environment Agency – Abu Dhabi) | Marine environment: environmental assessment, licensing of marine works and discharges, marine water-quality policy, protected species |
| Abu Dhabi Maritime / coastguard | Marine works notices, navigation safety in the approach |
| Partner lab ethics / animal care | Invertebrate welfare protocol for contained trials |

NUCLEAR-SAFETY INTERFACE — LEAD WITH THIS IF ASKED

High Tide sits 1–5 km seaward of the intakes, never inside them, and is designed so every failure leaves the plant exactly as it is today: air off, hose anchored with weak links that part before structures load, no floating parts, no chemicals until approved. It reduces the biological load on existing protection; it does not claim to replace any safety function.

# Six months to a pilot decision.

## Objectives

| ID | OBJECTIVE | MEASURED BY |
|---|---|---|
| O1 | Prove the wall | Surrogate deflection vs angle, current, air; air per metre needed |
| O2 | Prove the nudge | Flow-corrected directional swimming of adult Gulf *Catostylus* to nonenal (away) and Artemia water (toward) |
| O3 | Prove it adds up | Incremental exits of cues + bubbles over bubbles alone; welfare; non-target endpoints |
| O4 | Site it | Surveyed gate line, exits, current record, quoted section, EAD pre-application |

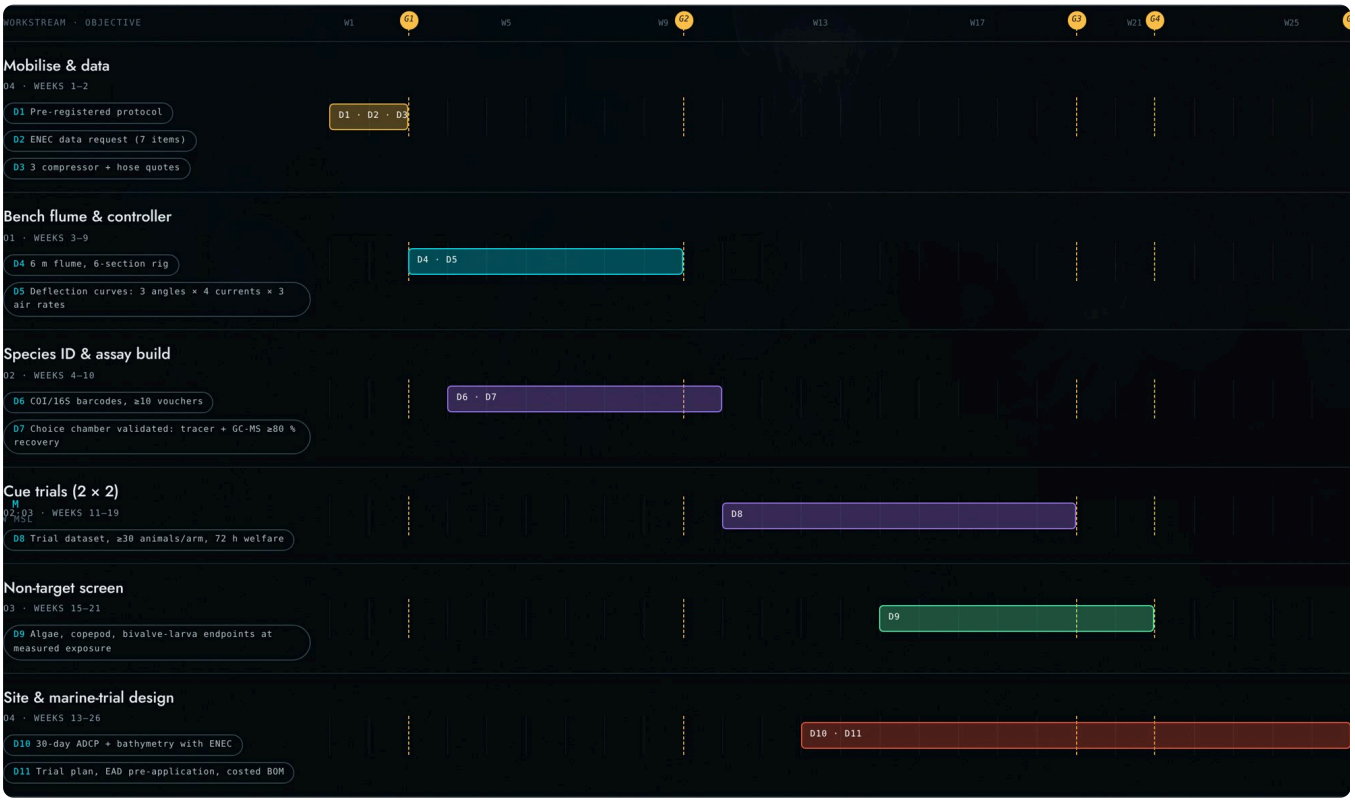



Fig. 9 — Workstreams, deliverables and gates.

## Deliverables

| ID | WEEK | DELIVERABLE | ACCEPTANCE |
|---|---|---|---|
| D1 | 2 | Pre-registered test protocol (e.g., OSF) | Hypotheses, endpoints, analysis fixed before data |
| D2 | 2 | ENEC data request (7 items) | Observation plane, currents, bloom load, warning-to-arrival, species, constraints, costs |
| D3 | 2 | Three compressor + hose quotes | Priced at 45 °C ambient duty |
| D4 | 5 | 6 m flume with 6-section controlled curtain | Repeatable flow ±10 % |
| D5 | 9 | Deflection curves: 3 angles × 4 currents × 3 air rates | ≥5 replicates per cell; tracer velocity maps |
| D6 | 8 | COI/16S barcodes of ≥10 vouchers | Species confirmed against Gulf references |
| D7 | 10 | Validated choice chamber | Tracer symmetry; GC-MS recovery ≥80 %; blank-arm preference vanishes with water only |
| D8 | 19 | 2 × 2 cue trial dataset | ≥30 independent animals per arm; blind tracking; 72 h welfare |

| | | | |
|---|---|---|---|
| D9 | 21 | Non-target screen | Algae, copepod, bivalve larvae at measured exposures |
| D10 | 20 | 30-day ADCP + bathymetry at the mouth (with ENEC) | Spring and neap captured |
| D11 | 26 | Marine trial plan + EAD pre-application + costed BOM | Zero-chemical, one recoverable 50 m section |

## Gates

| GATE | WEEK | PASS CRITERION | IF PASS | IF FAIL |
|---|---|---|---|---|
| G1 Ready to test | 2 | Protocol pre-registered; lab partner + ENEC data contact signed | Start builds | Hold, re-scope with ENEC |
| G2 The wall works | 9 | ≥70 % surrogate deflection at 0.2 m/s with ≤1.1 L/s per metre | Size the marine section | Double curtain or guard-only |
| G3 The nudge is real | 19 | ≥15 pp more verified exits than bubbles alone, $p < 0.05$, no excess injury at 72 h | Cues stay in design | Physical-only High Tide |
| G4 It's safe | 21 | No effect on algae, copepods, bivalve larvae at 10× design band | Containment design | Chemical layer dropped |
| G5 Pilot decision | 26 | Surveyed gate line; quoted 50 m section; EAD pre-application filed | Marine trial (zero chemicals) | Report and stop |

## Statistics planning example

For a binary first-choice test with 30 independent animals, 21/30 choosing the expected side gives exact two-sided binomial $p \approx 0.043$ vs 50:50; at a true preference of 75 %, that rule has ~80 % power. Primary analysis is flow-corrected lateral velocity, not first choice; animals are not re-counted across trials.

# What could go wrong.

| RISK | CONSEQUENCE | L | I | MITIGATION |
|---|---|---|---|---|
| Adults show no directional response to cues | Lose differentiator | M | M | Gate G3; physical-only High Tide still stands; test second candidate |
| Cue causes stress/immobility, not avoidance | Animals sink/drift in | M | H | Score pulsation, recovery; immobility counts as failure |
| Non-target toxicity at effective dose | Environmental harm | L–M | H | Gate G4; 10× margin; contained first; drop chemistry |
| Regulator declines chemical use | No stick/carrot at sea | M | M | Physical system designed to stand alone |
| Exits are not real exits (return flow) | Accumulation, returns | M | H | ADCP survey; return metric; carrot shuts off on returns |
| Waves/shamal destroy surface jet | Reduced deflection in storms | M | M | Operating envelope; fall back to screens |
| Biofouling/scaling clogs diffusers | Weak curtain | H | M | Purge cycles, slit sleeves, inspection, spares |
| Hose or anchor breaks loose | Debris near intake | L | H | Weak links, seaward siting, inspection, no floating parts |
| Compressor failure at peak | Gate off | M | M | N+1; fail-off; screens remain |
| Tracking misses at night/turbidity | Wrong sections fire | M | L–M | Sonar; fall back to fixed firing of routed line |
| Species misidentified | Wrong biology | M | M | Vouchers + barcoding (D6) |
| Hardware becomes polyp substrate | More local blooms | L–M | M | Smooth materials, cleaning schedule, monitoring |
| Security / access constraints at a nuclear site | Delays | H | M | Partner with ENEC teams; non-operational test area first |
| Cost higher than Class-5 range | No full build | M | M | Lean phase (one line + guard); V-gate option |
| Bloom load not as assumed (mass, timing) | Sizing error | M | L–M | D2 data request; sizing is area-driven |

L = likelihood, I = impact (L/M/H). Team's judgement.

# Where a sharp judge will push.

Each entry: the attack, our best answer, and what *not* to say. Rehearse these until any one of us can answer any of them in under 30 seconds.

PP01 **"Your chemical has never been shown to repel this species."**

**Answer:** Correct, and we say so on the page. We have a named molecule with measured enrichment around injured jellyfish (23×, $p < 0.001$) and a measured deterrent effect on another jellyfish; plus a separate, controlled side-preference experiment for food cues in medusae. Our first experiment tests directional response in identified Gulf adults. The physical corridor works without it.

Don't say: "it repels jellyfish", "it's proven", "it's a pheromone".

PP02 **"Miura's experiment was about a predator dropping food, not swimming away."**

**Answer:** Yes — it is a contact feeding-deterrence result in *Chrysaora pacifica*, using treated krill. It gives us a molecule worth testing, not a dose or a direction. That is why our endpoint is flow-corrected lateral swimming in a choice chamber, with uniform-concentration controls to separate agitation from direction.

PP03 **"You are putting a chemical into a nuclear plant's cooling water."**

**Answer:** Not in the first marine trial — zero release. If cues pass efficacy and safety gates, the band is microgram-per-litre, pulsed, and ~4 km from the pumps. Fully mixed into 690 $m^3/s$ it is ~0.26 µg/L — ~21,000× below the fish $LC_{50}$ screening estimate — and passes the condensers and outfall with everything else. EAD would have to approve it; we would never ask FANR to change a safety function.

Don't say: "it's harmless because it's a food flavour." Flavour status says nothing about marine toxicity.

PP04 **"Bubble curtains have been tried and failed (Lo 1991; fish studies)."**

**Answer:** Lo (1991) found a bubble barrier *alone* was not adequate in lab tests and recommended a net behind it — we keep the plant's screens and add a guard. Fish studies fail because fish are strong swimmers that habituate; jellyfish drift. Recent large-scale tests (C-CORE) show deflection up to 0.6 m/s with angled, multiple curtains, and Mowi saw up to 80 % fewer jellies inside a curtain. Our differences: routing, switching with the tide, verification, and siting where the current is weak.

PP05 **"Fixed bubbles gave ZERO intake entries — better than your controlled 78. Why not just run them?"**

**Answer:** Because zero entries came with 722 animals trapped and 7,875 return events — a pile-up in the approach that the next tide or a power blip delivers to the screens. Controlled sections cleared 266 vs 33. The metric that matters is animals that leave and stay gone.

PP06 **"Your model is a toy — 60 m, not CFD, uncalibrated."**

**Answer:** Agreed it's a design tool. It is deterministic, conserves every animal (balance error 0), includes failure scenarios where we lose (blocked exit, zero response, power loss), and reproduces exactly in the browser. It is also harsher than the real mouth (plume-to-flow ratio 2.5 vs ~13). We would calibrate it with flume data (D5) and ENEC currents (D10).

PP07 **"Where did 50,000 animals come from?"**

**Answer:** 200 t ÷ 4 kg. The 4 kg is our estimate for a 30–45 cm Catostylus by cube-scaling Indian *Catostylus* cf. *perezi* weights (mean 1.12 kg at 20.5 cm), cross-checked with *C. tagi* (~2.5 kg at 25 cm). Range 2–8 kg → 25,000–100,000. ENEC screen-wash records would replace it. Hardware sizing doesn't depend on it.

PP08 **"A 9 km bubble system is enormous. Isn't this over-engineered?"**

**Answer:** It's the honest consequence of the geometry: the basin mouth is 4.2 km wide, and that is where the physics favours us. But it's modular: phase 1 can be one line + guard (~5.3 km), a V-gate could halve hose, and the pilot is one 50 m section. Even full scale draws <0.02 % of the plant's output and pays back after 2–11 avoided unit-days.

PP09 **"How do you know where the exits lead? Maybe they come back next tide."**

**Answer:** We don't yet — that's why "returns" is a first-class metric and why D10 (30-day ADCP) is on the plan. Exit N releases outside the breakwater on the flood, where the coastal current runs away from the intake basin; Exit W releases along the open shore on the ebb. If the record shows strong return, we move the gate or exits.

PP10 **"Is the species even *C. mosaicus*?"**

**Answer:** Possibly not — Gulf records point to *C. perezi*. We use ENEC's label for sizing and make species ID our first biology deliverable (D6). Response curves are estimated per confirmed species.

PP11 **"What happens in a shamal, at night, in murky water?"**

**Answer:** Night/turbidity: imaging sonar plus fallback to firing the routed line in full. Storms: waves degrade surface jets; the operating envelope says "screens carry the load above X wave height" — X to be agreed with ENEC from trial data. The system fails off, never on.

PP12 **"Could your hardware damage the plant?"**

**Answer:** Nothing is in or attached to the intake structures. Hoses are 1–5 km seaward, ballasted, on weak-link tethers that part before any load builds; no floating parts. A loose hose sinks. The guard line is the nearest element and is still outside the channel.

PP13 **"Why not just collect and sell them? Jellyfish are food."**

**Answer:** A collection fishery could pair with our exits (animals concentrated at one point is a gift to a collector), but ENEC asked to minimise manual work and protect ecosystems. *Catostylus* is fished in Asia/Australia; UAE has no jellyfish fishery and processing needs alum/salt and a market. We route; ENEC could add collection at the exit later.

PP14 **"Why should a jellyfish care about a 5 µg/L band when it's being carried at 3 cm/s?"**

**Answer:** That's exactly why we put cues at the mouth: there the carrying current is about the animal's own speed, so a sustained turn changes the outcome. Closer in, cues can't win and we don't rely on them. Whether 5 µg/L is enough is unknown — 5 µg/L is a sizing bracket, not a claim.

PP15 **"How do you separate chemistry from currents in your lab test?"**

**Answer:** Balanced flows measured in both arms, side-swaps, dye and analytical chemistry to map the plume, water-only runs where preference must vanish, uniform-concentration runs to separate activity from direction, swimming velocity = tracked motion – local water velocity, blind scoring, pre-registration.

PP16 **"What about dugongs, turtles, dolphins?"**

**Answer:** No nets (no entanglement — the leading dugong mortality cause), low-noise air (bubble curtains are used to reduce noise), no chemicals in the first marine trial, observation protocol during trials, and siting away from seagrass after a habitat survey. Marawah is ~100 km away; Barakah's basin still gets surveyed.

PP17 **"What's actually novel? Bubbles exist. The molecule exists."**

**Answer:** The system: controlled, tide-switching, verified routing; siting by the pull-vs-swimmer rule; and a 2 × 2 test that measures whether behaviour cues add value on top of physics. Nobody deploys behavioural steering for jellyfish at plants today.

PP18 **"Your costs are made up."**

**Answer:** They are labelled Class-5 with sources and assumptions shown (per-metre curtain benchmark, compressor cost per kW, sensor allowances). Deliverable D3 replaces them with quotes in two weeks. The pilot ladder costs are itemised.

PP19 **"You're students. Can you build this?"**

**Answer:** We can build and test the bench system and controller ourselves (robotics, embedded control, software, molecular biology on the team). The biology runs with a partner lab under animal-care rules; marine works with ENEC's contractors. The plan reflects that split.

PP20 **"Barakah doesn't have a jellyfish problem — the screens cope."**

**Answer:** Then this is insurance: Gravelines also coped for 30 years before losing four reactors in 2025 and three in 2026. Warming seas push blooms up. And coping today costs crews, disposal and dead animals — exactly what ENEC asked to reduce.

# Every question we could think of.

Difficulty: 1 easy · 2 probing · 3 killer. "A" = answer; "If pushed" = follow-up.

## A · The problem and the brief

**1 In one sentence, what is High Tide?**

A controlled bubble corridor, plus chemical push and pull cues, that routes jellyfish away from Barakah's intakes after ENEC's alert and counts what clears.

**1 Why not early detection?**

ENEC already has it and the brief excludes it. We use their alert as our trigger.

**2 What does "minimise manual intervention" mean in your design?**

No boats, crews or nets in normal operation; operator acknowledges alerts and all-clear; maintenance is scheduled, not reactive.

If pushed: maintenance — diver inspection of hose annually, compressor service, culture care weekly.

**2 How do you define success?**

Tonnes reaching the screens per event down; verified clears up; returns low; no harm to jellyfish or non-targets; no new manual work.

**2 Where does 200 t/day come from?**

ENEC mentors cited it as a historic peak daily load on 30 Oct. We need timestamped records to size properly (D2).

**2 Is 200 t at the screens or offshore?**

We understood it as load handled at the plant. If it's an offshore bloom estimate, the screen load is lower and our margins larger. We ask in D2.

**1 Why focus on Barakah and not generic infrastructure?**

The brief's sponsor is ENEC; a specific site forces real numbers. The approach generalises to desal plants and other intakes.

**2 Has Barakah ever tripped from jellyfish?**

Not that we've seen publicly reported. We frame High Tide as insurance and workload reduction.

## B · Biology

**1 What is Catostylus mosaicus?**

A rhizostome scyphozoan ("blue blubber"), domed bell, eight oral arms, no long tentacles, mild sting, forms large blooms.

**3 Are you sure the Barakah species is C. mosaicus?**

No. Gulf records point to C. perezi. We confirm with vouchers and COI/16S barcoding before biology claims.

**2 How fast does it swim?**

~36 mm/s in lab for 2–5 cm animals (Neil & Askew 2018). Adults unmeasured; probably faster absolutely.

**2 Can jellyfish even sense chemicals?**

Yes in general: Hydra feeding reflex to glutathione (10 µM half-max); Aurelia side preference for Artemia water (Arai 1991); planula settlement responds to predator cues (2026).

**2 Do they have a brain?**

No; a nerve net plus rhopalia (sensory structures with gravity sensors and often light-sensitive spots).

### 2 Why do blooms happen?

Polyps on hard surfaces release many ephyrae when triggered; warmth, nutrients, fewer predators/competitors and more hard substrate raise bloom size.

### 3 Do Barakah's breakwaters breed the jellyfish?

Possible — artificial hard substrate can host polyps. Unknown locally; worth surveying with EAD. High Tide hardware is smooth and cleaned so it doesn't add habitat.

### 2 When is the season?

UAE survey reports aggregations Oct–Nov on the Abu Dhabi coast; present year-round.

### 2 Will bubbles hurt them?

No contact or pressure change; worst plausible is bubbles caught under the bell — scored in trials.

### 2 Do jellyfish swim actively against currents?

Yes — adult Rhopilema and Rhizostoma swam faster against the current in flumes (Malul 2019). It's why we control for rheotaxis.

### 3 Could steering them away just move the problem to another facility?

Exits release along open coast with natural currents; the next large intake (Ruwais) is ~50 km away. We would share verification data with neighbours; animals are alive and disperse.

### 2 Can they habituate to the cue?

Possible. The assay includes repeated exposure and recovery periods; pulsing (40 % duty) reduces constant exposure.

### 2 Size dependence?

Response factor in the model scales weakly with bell size; trials test a size distribution.

## C · The stick — chemistry and evidence

### 1 What is (E)-2-nonenal?

A 9-carbon unsaturated aldehyde ($C_9H_{16}O$), the "cucumber aldehyde"; a flavour/fragrance ingredient (FEMA 3213).

### 2 Explain 23×, n = 10, p = 0.00094 to me as a non-scientist.

Water around an injured jellyfish had 23 times more of it than around an unhurt one; ten samples each; if there were no real difference, a gap that big would show up by chance less than once in a thousand tries.

### 3 Is it a pheromone?

No. It's a defence chemical released by injured skin (allomone) that deterred a predator. A pheromone signals within a species; nobody has shown that here.

### 3 Which species released it, and which reacted?

Released by the hydromedusa Spirocodon saltatrix; the predatory sea nettle Chrysaora pacifica released treated krill faster. Neither is our species.

### 2 What concentrations worked?

Krill pretreated at 50 % and 5 % formulations (0.5 % n.s.). These are surface treatments, not water concentrations — they can't set our dose.

### 2 So where does 5 µg/L come from?

It's a sizing bracket (1–10 µg/L) chosen to sit far below screening toxicity values and within practical dosing; the assay replaces it with a measured effective concentration.

### 2 How soluble is it? How do you dose it?

Est. ~205 mg/L; we pre-dilute to 100 mg/L in filtered seawater (~16 $m^3$/h carrier at the mid case) and release through a 600 m diffuser.

### 2 How long does it last in water?

It volatilises and oxidises; persistence is short but must be measured in Gulf water (part of D7).

### 3 Is it toxic?

Irritant at handling strengths. ECOSAR predictions: fish $LC_{50}$ ~5.5 mg/L, algae $EC_{50}$ ~8 mg/L — screening estimates, not safety thresholds. Our band is 5 µg/L (≈1,000× lower); at pumps ~0.26 µg/L. G4 tests real endpoints.

### 2 Does it occur naturally in the sea?

It's reported from brown algae and dinoflagellate cultures, so background levels exist — we measure them.

### 2 Cost?

Flavour-grade bulk ~USD 300–1,000/kg (assumption) → ~AED 8.6–28.8k per 12 h event at the mid case.

### 2 Why not 1-octen-3-ol or (E)-2-octen-1-ol, which worked at 0.5 %?

They're backup candidates. Octenol wasn't found in injured-water samples, so nonenal has both a natural-release and a behaviour link. The assay tests a second molecule.

### 3 Couldn't the cue make them sink or stop swimming — and drift into the intake?

That's a failure condition in our protocol: we score pulsation and immobility, and accept only sustained directional swimming with recovery.

### 2 What if it attracts instead of repels?

Then it becomes a candidate carrot. The assay is two-sided.

### 2 Why not ammonia, which attracted Aurelia?

Ammonia toxicity rises with temperature and pH — unacceptable in warm Gulf water near an intake.

## D · The carrot

### 2 Why brine shrimp?

A standard, safe, cultured prey whose conditioned water attracted medusae in a controlled two-inflow test.

### 2 How much Artemia?

5–54 kg standing culture producing 0.5–5.4 $m^3$/h of conditioned water; the shrimp are reused.

### 3 Aren't you just attracting more jellyfish to the plant?

Only at the exit currently clearing, outside the gate; off when returns rise. A pull toward a door that's open.

### 2 Nutrients and water quality?

Filtered water, small volumes vs 690 $m^3$/s; monitored at the exit sonde.

### 2 Biosecurity?

Certified cysts; no live shrimp released; culture on shore.

## E · Bubble corridor engineering

### 1 How does a bubble curtain deflect?

Rising bubbles lift water; at the surface it spreads sideways at 0.3–0.4 m/s, holding drifting animals back while the background flow slides them along the line.

### 2 Where do 0.30–0.43 m/s come from?

Bulson's relation with 0.55–1.67 L/s·m (Beelen basin data) at 6 m depth.

### 2 Why 30°?

Angled curtains deflect better at higher currents (C-CORE tests 5/30/45°); the angle also sets the slide toward an exit. Optimal angle is a D5 output.

### 2 Why 50 m sections?

Matches tracking resolution and keeps valve count (~194) manageable; pilot uses 10 m.

### 2 How do you switch sides?

Two physical lines (A and B); the controller fires one based on ADCP current direction, 10 s after a reversal.

### 3 Does a bubble curtain work in 0.3 m/s channel flow?

Less well; that's why the guard is double and sized for a 0.6 m/s jet, and why the main gate sits where flow is 0.03 m/s.

### 2 Fouling?

Expected; slit sleeves, purge cycles, trickle flow, inspections, spares.

### 2 Heat?

Compressors sized and cooled for 45 °C ambient; aftercoolers; shaded enclosure.

### 2 Why oil-free?

No oil sheen in the sea; standard for marine bubble systems.

### 2 Anchoring?

60–100 kg clumps every 5 m, weak-link tethers; Diablo Canyon commissioned anchoring studies for its curtain — a known design problem.

### 2 Waves?

Degrade surface jets; envelope-limited operation; screens carry storms.

### 2 Navigation?

Seabed hose, no surface floats; the basin is not the navigation channel; marine notices for works.

### 2 Does the curtain block cooling water?

No — water passes through bubbles; only surface layers are redirected. Net intake flow is unchanged.

### 3 Couldn't the upwelling pull deeper jellyfish to the surface and then into the intake?

Upwelling brings them to the surface where the sideways jet is strongest; that's the intended capture zone of the curtain. Deep-drifting animals are a known gap — sonar tracks depth; double curtain at the guard.

### 2 Why not water jets?

Pumping water costs more energy per unit of surface current and can entrain animals; air is simpler and benign.

### 2 Why not a net?

Nets clog, block water, kill animals and entangle dugongs/turtles; high maintenance.

## F · Sensing and control

### 2 How do you track jellyfish?

EO/IR cameras on masts (surface bells by day); imaging sonar at the gate (night/turbid). Tracks feed the 13 m trigger.

**2 Can cameras see translucent jellyfish?**

Blue blubbers are visible from above in calm water; glare and turbidity are limits — hence sonar and the fallback of firing the full routed line.

**2 What if tracking fails?**

Degraded mode: fire the routed line fully (equivalent to "fixed" in that tide); log it.

**2 How do you know the current direction?**

ADCPs at gate, basin, channel, plus tide tables; switch after measured reversal.

**2 Human in the loop?**

Operator acknowledges alert, can override; routine section switching is automatic.

**2 Cyber security on a nuclear site?**

Controller isolated from plant control networks; data out via one-way link; follows ENEC IT/OT policy.

**2 Could AI improve it?**

Track classification (jellyfish vs debris), arrival forecasting from ENEC's alert, and adaptive section scheduling are natural next steps — data first.

## G · Model and simulation

**2 What exactly does the model compute?**

Positions of up to 900 animals every 0.25 s under along-flow, cross-flow, intake pull, dispersion and curtain/cue terms; tallies outcomes.

**3 Is it validated?**

Not against field data. It's verified (mass balance 0, reproducible) and used comparatively. D5 and D10 calibrate it.

**3 Did you tune it to look good?**

It includes scenarios where we fail (blocked exit, zero response, power loss, return current), all three seeds archived, CSV published.

**2 Why three seeds?**

To show results aren't a single lucky draw; they vary by a few percent.

**2 What's the cue layer assumption?**

A swim bias capped at the animal's own speed (36 mm/s) toward the exit / away from the intake plume. Nothing stronger.

**3 Your model's approach flow is 0.22 m/s but your site says 0.03 m/s. Which is it?**

Both are right for different places: the model is a harsh generic approach (closer to the throat); the gate sits where flow is ~0.03 m/s, so real conditions favour the curtain more than the model.

**2 Why 900 animals in 15 minutes?**

1/s per 60 m strip — much denser per metre than Barakah's peak (~1.7/s across 4.2 km); a stress test.

**2 Why do some animals leave by "another boundary"?**

They drift out of the domain sideways/backward; we report them separately rather than call them clears.

**2 Is the 3D simulation real?**

The positions and counters come from the engine unchanged; the scenery is illustrative.

## H · Site

**2 How did you get the geometry?**

Traced from public satellite imagery (Esri World Imagery) with a lat/long grid; checked by overlay at several zooms. Indicative only.

**2 Where are the four intakes?**

Four stepped structures along the plant frontage, ~52.226–52.238° E, facing a ~250–300 m dredged channel; numbered 1–4 west→east by us.

**2 Where is the discharge?**

East of the intakes; warm water exits through an eastern channel between breakwaters toward the north-east, separated from the intake basin by the breakwater.

**3 What's the point of no return and how robust is it?**

Where mean intake pull = swim speed: ~4.3 km out. It depends on assumed depths (5.5–8.5 m) and swim speed; ±1 m depth moves it a few hundred metres. ADCP and bathymetry fix it.

**2 Why the mouth and not closer?**

Closer in, water outruns the animal and exits dead-end at the breakwater; at the mouth both exits are open sea.

**2 What is that straight line across the basin?**

An unverified linear feature in imagery — maybe a boom or pipe. We ask ENEC; not part of our design.

**2 Will the gate interfere with ENEC's marine monitoring or the artificial reef?**

Sited after a habitat survey; reef areas avoided; we'd coordinate with ENEC's environmental team.

**2 Why is the shore station where it is?**

Closest land with power near the west of the plant; final location by ENEC.

**2 Could you use the breakwater for cables and pipes?**

Yes — feeders to Exit N are routed along the breakwater crest in our estimate.

## I · Numbers and economics

**2 How much power?**

913 kW worst case (0.016 % of 5.6 GW); ~442 kW typical; 5.8 MWh per 12 h event.

**2 How much air?**

~290–300 $m^3$/min free air per gate line at full; guard 170 $m^3$/min.

**3 How much does it cost?**

Pilot: AED 2–9k bench, 36–125k biology, 106–486k marine trial. Full scale: AED 18–67 M Class-5.

**3 What's the ROI?**

Payback after 2–11 avoided unit-days (unit-day ≈ AED 6–10 M), plus crew, disposal and screen wear. We present it as insurance.

**2 Why USD 50–80/MWh?**

A neutral range for wholesale value; Barakah's PPA price isn't public.

**2 What about OPEX?**

Energy ~AED 1.7k/event; chemicals 8.6–28.8k/event if approved; maintenance (compressors, diver inspection, sleeves) annual — to quote in D3.

**2 How does this compare to EDF's boats?**

Seven boats 24/7 means crews, fuel and weather limits every event; our marginal event cost is power and (maybe) cue chemicals.

**2 Is the CAPEX dominated by hose?**

Yes (~9.7 km). Lean options: one line + guard, or a V-gate.

## J · Environment and regulation

**2 Who regulates this?**

EAD (marine environment, discharges, works licensing), FANR (nuclear safety interface), ENEC (site owner/operator).

**3 Does FANR need to approve?**

We avoid touching safety-related structures; ENEC would decide if any interface review is needed. We design to fail-off and stay out of the intake.

**2 Noise?**

Bubble curtains emit low broadband noise and are themselves used to block construction noise; measure in trial.

**2 Effect on fish?**

Most fish habituate; no physical harm; monitored.

**2 Seagrass and corals?**

Habitat survey before siting; avoid sensitive areas; anchors small.

**3 What if the chemical bioaccumulates?**

log P ~3.2 suggests modest potential; we measure in the non-target programme. If it accumulates, the chemical layer is dropped.

**2 Ethics of moving wild animals?**

We move them without contact, alive, to open sea — gentler than screens (death) or shredders.

**2 Carbon footprint?**

~5.8 MWh/event of low-carbon electricity from the plant itself; negligible.

## K · Plan, team, feasibility

**2 What happens in the next 2 weeks?**

Pre-register protocol, send the 7-item data request, get three quotes, sign a lab partner (G1).

**2 Which lab partner?**

A UAE marine lab with animal care and analytical chemistry (candidates to discuss with ENEC/EAD: university marine labs in Abu Dhabi and Al Ain). Not yet signed.

**2 What if G3 fails?**

We ship the physical system: controlled, verified routing still beats fixed curtains in our model.

**2 What is your prototype at Demo Day?**

The live model, the site plan, and the bench concept with inert surrogates and tracer — no live animals or chemicals at the booth.

**2 Who does what in the team?**

Syed — lead/robotics; Usman — AI and molecular biology; Randall — embedded systems and controls; Mustafa — software. (Adjust to match the team's own roles.)

### 2 Biggest technical risk?

Cue response in adults (G3). Biggest practical risk: access and data at a nuclear site.

### 3 If we gave you AED 500k tomorrow, what would you have in six months?

D1–D11: flume curves, species ID, a validated choice chamber, a cue dataset with non-target screen, a surveyed gate line and a costed, permitted-ready 50 m trial.

## L · Curveballs

### 2 Why not cool the water or use temperature?

Heating or cooling the intake flow is a scale mismatch: 1 °C on 690 $m^3/s$ ≈ 2.8 GW thermal. Temperature stays a test factor, not a tool.

### 2 Why not light?

Juvenile moon jellies preferred bright compartments in one study; adult Gulf response unknown, and light affects fish and turtles. A possible future comparator.

### 2 Why not sound?

Jellyfish lack swim bladders; acoustic deterrence works on fish, not on gelatinous drifters.

### 2 Why not move the intake deeper?

Major civil works at an operating nuclear plant — out of scope; EDF is doing it only for new EPR2 units.

### 2 Why not kill polyps?

Biocides near reefs and a nuclear intake are worse than the problem; and polyp locations are unknown.

### 2 Could eDNA measure the bloom?

Good for presence, poor for abundance (concentration ≠ biomass); can't populate tonnage counters.

### 3 What if ENEC already has a bubble curtain?

Then the controller, routing, verification and cues upgrade it — the same logic runs on existing hose.

### 3 What's the one result that would kill the project?

If, in the flume, surrogate deflection at Barakah-like currents needs far more air than estimated (G2 fail) and ADCP shows strong return flow at the exits — then routing has nowhere to go. We'd say so.

### 2 What would make you more confident tomorrow?

ENEC's screen-load timestamps and one month of current data at the mouth.

### 2 Why the name High Tide?

Blooms ride the tide in; we use the tide to send them out.

### 2 Is anything on your website made up?

No data is invented. Illustrative elements (3D scenery, aquarium footage, procedural jellyfish) are labelled; assumptions are tagged; model outputs are archived.

### 2 How does this generalise?

Any seawater intake: desal plants (Gulf has many), LNG, data centres, aquaculture pens.

### 2 Could the jellyfish get stuck at the exits?

That shows as "retained/returned"; carrot stops, routing changes, and the exit location is reconsidered.

### 2 Do you need permits for the bench test?

No marine permits for inert surrogates; animal-care approval at the partner lab for live animals.

### 2 What about other jellyfish species in a mixed bloom?

The curtain is species-agnostic; cues may not be. Each confirmed species gets its own response estimate.

### 2 Why should ENEC trust students with this?

Because the plan puts every claim behind a measurable gate and asks ENEC to judge on data, starting with a cheap bench test.

### 3 What did you get wrong in your first version?

We led with the load, the map was a regional picture, the cue statistics weren't explained and the plan lacked deliverables. v2 fixes all four and adds the pull-vs-swimmer siting rule.

# Words to use precisely.

| | |
|---|---|
| **ADCP** | Acoustic Doppler current profiler — measures current speed and direction through the water column. |
| **Allomone / kairomone / pheromone** | Chemical signals benefiting the emitter / the receiver across species / within one species. |
| **APR-1400** | Korean advanced pressurised-water reactor, 1,400 MWe class; Barakah has four. |
| **Bubble curtain** | Line of rising air bubbles from a seabed diffuser; creates upwelling and a horizontal surface current. |
| **Class-5 estimate** | Concept-stage cost estimate, typical accuracy –50 % to +100 % (AACE). |
| **ECOSAR** | US EPA structure-based model predicting aquatic toxicity — screening only. |
| **Ephyra / polyp / strobilation** | Juvenile medusa / attached stage / budding of ephyrae from a polyp. |
| **$LC_{50}$ / $EC_{50}$** | Concentration killing / affecting 50 % of test organisms in a set time. |
| **Mesoglea** | Jelly-like middle layer of a jellyfish's body. |
| **Once-through cooling** | Seawater pumped through condensers once and returned to the sea. |
| **p-value** | Probability of seeing a difference at least this large if there were no real effect. |
| **Rheotaxis** | Orienting to water current. |
| **Rhizostome** | Order of scyphozoan jellyfish with fused oral arms and no marginal tentacles. |
| **Rhopalium** | Sensory structure on the bell margin with statocyst and often ocelli. |
| **Ultimate heat sink** | The final destination of plant heat — at Barakah, the Gulf. |
| **Voucher** | A preserved specimen or tissue sample kept to verify species identity. |

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